MIL-F-46851B
20 November 1980
Superseding
MIL-F-46851A
9 August 1968

MILITARY SPECIFICATION

FANS, GUIDED MISSILE, GROUND SUPPORT EQUIPMENT, GENERAL SPECIFICATION FOR

1. SCOPE

1.1 Scope. This specification covers the general requirements for three types of general purpose, air circulation fans to be used within the ground support equipment of a guided missile system.

1.2 Classification. Fans covered by this specification shall be of the following types as specified (see 6.2):

Type I - Centrifugal (blower) fans

Type II - Vane axial fans

Type III - Propeller fans

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, For Carton Sealing

PPP-B-601 - Box, Wood, Cleated-Plywood

PPP-B-636 - Box, Fiberboard

Beneficial comments (Recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to; Commander, US Army Missile Command, ATTN: DRSMI-RSDS, Redstone Arsenal, AL 35898 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1450

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PPP-C-850 - Cushioning Material, Polystyrene Expanding, Resilent for Packaging Uses

Military

MIL-P-116 - Preservation, Method of

MIL-B-121 - Barrier Material, Greaseproof, Waterproof, Flexible

MIL-T-152 - Treatment, Moisture and Fungus-Resistance, of Communications, Electronic, and Associated Electrical Equipment.

MIL-P-514 - Plates, Identification, Instruction and Marking Blank

MIL-D-3464 - Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification

MIL-A-8625 - Anodic Coatings, for Aluminum Alloys

MIL-C-16173 - Corrosion Preventive Compound, Solvent Cutback, Cold Application

STANDARDS

Military

MIL-STD-105 - Sampling Procedure and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-130 - Identification Marking of U.S. Military Property

MIL-STD-171 - Finishing of Metal and Wood Surfaces

MIL-STD-202 - Test Method for Electronic and Electronic Component Parts

MIL-STD-461 - Electromagnetic Interference Characteristic Requirements for Equipment

MIL-STD-810 - Environmental Test Methods

MIL-STD-889 - Dissimilar Metals

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MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

MIL-STD-1472 - Human Engineering Design Criteria for Military Systems Equipment and Facilities

MS2000-3 - Indicator, Humidity, Card, Three Spot, Impregated Areas (Cobaltous Chloride)

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the processing activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

Air Moving and Conditioning Association, Inc. - Bulletin No. 210
Standard Test Code for Air Moving Devices

(Application for copies should be addressed to the Air Moving and Condition Association, Inc., 2159 Guardian Bldg., Detroit, Michigan).

Air Filter Institute

Code for testing air cleaning devices used in general ventiliation.

(Application for copies should be addressed to the Air Filter Institute, 215 Central Avenue, Louisville, Ky).

3. REQUIREMENTS

3.1 Precedence. Detail requirements applicable to the individual type fans shall be as specified on the detail drawing. In the event of conflicting requirements between this specification and the detail drawing, the latter shall govern.

3.2 Preproduction model. Unless otherwise specified (see 4.3.1.1 and 6.2), preproduction models shall meet the requirement specified herein. After approval of the model, no changes shall be made to the fan without written approval of the contracting officer.

3.3 Design and construction.

3.3.1 Parts, materials and processes. Parts, materials and processes shall be as specified in this specification and the application detail drawing. Acceptance or approval of any constituent materials or part shall not be construed as acceptance of the fan.

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3.3.1.1 Metals. Metals shall be corrosion-resistant type, or shall be plated or treated to resist corrosion.

3.3.1.1.1 Dissimilar metals. When dissimilar metals contact each other, protection against electrolysis and corrosion shall be provided. Dissimilar metals shall be as defined in MIL-STD-889.

3.3.1.2 Aluminum alloy parts. Where aluminum alloy parts are anodized, the anodic treatment shall be as specified in MIL-A-8625.

3.3.2 Weight and dimensions. The weight and dimensions of the fan shall be as specified on the detail drawings.

3.3.3 Finishes. Exterior finishes for the fan shall be as specified on the detail drawings and shall conform to MIL-STD-171.

3.3.4 Marking. Each fan shall be identified by means of a durable nameplate as specified in MIL-P-514, and shall include the supplier's name, trademark, part number, and date manufactured. The nameplate shall be clearly and permanently marked as specified in MIL-STD-130..

3.3.5 Fungus treatment. Parts and materials shall be treated with fungicidal material to the greatest extent. With the exception of parts and materials which are in hermetically sealed cases, any parts or material not inert to fungus growth shall be individually treated with fungicidal material. No damage shall result from treatment of fungicidal coating. The application of fungicidal treatment shall be in accordance with type II MIL-T-152.

3.3.6 Age limitation. The fan shall not be acceptable if manufactured more than 36 months prior to delivery. The date of manufacture shall be stamped on the fan.

3.3.7 Interchangeability. Uniformity of shapes, dimensions, and performance shall permit interchangeability of replaceable parts and the complete fan of the same part number.

3.4 Performance characteristics.

3.4.1 Fan and motor characteristics. The fan and motor characteristics for air flow and power requirements shall be as specified on the detail drawings.

3.4.2 Dielectric withstanding voltage. Unless otherwise specified on the detail drawing, the fan shall not exhibit arching or insulation breakdown when subjected to 1000, plus or minus 100 volts, alternating current (vac), 60 hertz Hz between the insulated terminals tied together and the motor housing, for 30, plus or minus 15 seconds.

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3.4.3 Insulation resistance. The fan motor shall have a minimum insulation resistance as specified on the detail drawing when subject to 500 plus or minus 50 volts direct current (vdc), between the insulated terminals tied together and the motor housing, for two minutes.

3.4.4 Power factor. The motors shall maintain, or be corrected to maintain at least .80 power factor.

3.5 Environmental and service conditions.

3.5.1 High temperature. The fan shall withstand a temperature of $71^{\circ} \pm 2^{\circ}$ Centigrade (C) ($160 \pm 4^{\circ}$ Fahrenheit (F) for not less than six hours without evidence of damage, and shall meet the requirements of 3.4.1.

3.5.2 Low temperature. The fan shall withstand temperature of $-54^{\circ} \pm 2^{\circ}$ C ($-65^{\circ} \pm 4^{\circ}$ F) for not less than six hours without evidence of damage and shall meet the requirement of 3.4.1.

3.5.3 Temperature cycling. The fan shall withstand temperature of -55° to $+ 71^{\circ}$ C (-67° to $+ 160^{\circ}$ F) for periods of 10 to 30 minutes, cycled five times without evidence of damage, and shall meet the requirements of 3.4.1.

3.5.4 Humidity. The fan shall withstand exposure to a relative humidity of 90 to 95 percent at $40^{\circ} \pm 2^{\circ}$ C ($104^{\circ} \pm 4^{\circ}$ F) for 240 hours without evidence of damage and shall meet the fan and motor characteristics of 3.4.1 within five seconds. The fan motor shall also meet the insulation resistance requirements as specified in 3.4.3.

3.5.5 Salt spray. The fan shall withstand exposure to five percent salt solution (NaCl) fogging for 48 hours without evidence of damage at a temperature of 35° C to 1° or -2° (95° F + 2° or -3°). The fans shall also meet the requirements of paragraph 3.3.4, 3.4.1 and 3.4.3.

3.5.6 Barometric pressure. The fan shall withstand an atmospheric pressure from 17 pounds per square inch absolute (psia) to a pressure equivalent to 10,000 feet altitude without evidence of damage, and shall meet the requirements of 3.4.2 and 3.4.3.

3.5.7 Shock. The fan shall withstand two impact shocks of 15 gravity units (g) along each two mutually perpendicular axes for a total of eight shocks, with each shock impulse having a time duration of 11 plus or minus one milliseconds without evidence of physical damage, and shall meet the requirements of 3.4.1, 3.4.2 and 3.4.3.

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3.5.8 Vibration. The fan shall withstand harmonic motion having an amplitude of either 0.06 inch double amplitude or 10 g, whichever is less, from 5 to 500 to 5 Hz in 15 minutes, except that the double amplitude shall be 0.125 inch for vibrations below 40 Hz. Fans shall be vibrated six times in each of three mutually perpendicular directions. The fans shall exhibit no damage and shall meet the requirements of 3.4.1, 3.4.2 and 3.4.3.

3.5.9 Fungus. The fan shall meet the requirement of MIL-STD-810 method 508, without evidence of harmful fungus growth and shall meet the requirements of 3.3.4 and 3.4.1.

3.5.10 Operational life. The fan shall withstand the time and temperature life requirements as specified on the detail drawings, and shall meet the requirements of 3.4.1, 3.4.2 and 3.4.3.

3.5.11 Radio interference reduction. The radio interference of each fan shall comply with requirements of MIL-STD-461.

3.5.12 Noise. The fan developed under this specification shall meet the noise requirements of MIL-STD-1472 Noise Criteria (NC) curves for speech communication NC 45 to NC 60, when measured at a distance of not more than 24 inches in any direction from the fan, when operating under any design load (pressure, head, etc.) in unconfined free air between sea level and 1500 feet above sea level.

3.6 Workmanship. There shall be no cracks, dents, scratches, burrs, sharp edges, loose solder, loose electrical connections, loose paint, or any other evidence of poor workmanship that will render the fan unsuitable for the purpose intended. The fan shall be clean and free of foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspection are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Test equipment. Test equipment and inspection facilities shall be of sufficient accuracy and quality to permit performance of the inspections and tests specified herein. The supplier shall establish adequate calibration of test equipment to the satisfaction of the Government. Records of test equipment calibration shall be maintained by the supplier for the duration of the contract or purchase order. The specific test equipment used for pilot model tests of fans shall also be documented. The accuracy of test instruments shall be verified and traceable to the National Bureau of Standards.

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4.2 Measurements.

4.2.1 Recalibration errors. If an instrument is scheduled for recalibration during a test period, the recalibration date shall be waived until completion of testing. If upon recalibration, instrument error exceeds specifications, a note shall be added to all test data collected prior to recalibration indicating percentage and direction of error deviation.

4.2.1.1 Measurement accumulative error. Total accumulative error for any one test shall not exceed the specified tolerance.

4.2.1.2 Stability of instrumentation. Stability of instrumentation shall be assured by preconditioning measurement instrument for a minimum of two hours prior to performing any measurements.

4.2.2 Standard condition. Unless otherwise specified (see 6.2), all tests required by this specification shall be made under the following standard conditions:

(a) Temperature $25^{\circ} \pm 2^{\circ}$ C ($77^{\circ} \pm 4^{\circ}$ F)

(b) Relative humidity - 70 percent maximum

(c) Barometric pressure - Local standard

(d) Standard air - rating of the fan shall be based on standard air. Standard air shall be considered as air at a temperature of 21° C (70° F), a barometric pressure of 29.92 inches of mercury, and a density of 0.075 pounds per cubic foot.

4.2.3 Repeatability. When it is necessary to repeat a measurement for comparison purposes, the repeated measurement shall be made at within two degrees C (plus or minus 4 degrees F) of the temperature at which the original measurement was made. Instrumentation, measurements, and nondestructive tests shall be conducted in a manner to be repeatable.

4.2.4 Performance measurement tolerances. When a test frequency is specified with no tolerance, the frequency used shall be within ten percent of the value specified. Test frequencies shall contain not more than one percent total harmonic distortion.

4.2.5 Environmental measurement tolerances. Unless otherwise specified (see 6.2), the maximum allowable tolerance on environmental measurements shall be as follows:

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Environment	Tolerance
(a) Temperature	Plus or minus two degrees C (plus or minus four degrees F)
(b) Altitude (feet)	Plus or minus five percent
(c) Relative humidity	Plus or minus zero percent
(d) Vibration amplitude	Plus or minus ten percent
(e) Vibration frequency	Plus or minus five percent
(f) Shock amplitude	Plus or minus ten percent
(g) Shock time	Plus or minus one millisecond
(h) Static pressure	Plus or minus 0.1 inch of water
(i) Additional tolerance	Plus or minus ten percent

4.2.6 Repetitive measurements. Insofar as it is practicable, measurements which are repetitive shall not be made unless otherwise specified on the detail drawing. The measurements taken after specific test shall be considered to be the initial measurements for the next test. Maximum elapsed time between tests, insofar as repetition measurements are concerned, shall not exceed 72 hours.

4.2.7 Temperature stabilization. For the purpose of this specification, a temperature chamber and the specimens contained therein shall be considered to have been stabilized at a specific temperature of two consecutive measurements, taken 15 minutes ± 1 minute apart on the largest mass of the part, are within ± 2 °C (± 4 °F) of the specific temperature.

4.3 Classification of tests. Testing of the fan shall be classified as follows:

(a) Preproduction tests (see 4.3.1)

(b) Acceptance inspection and tests (see 4.3.2)

4.3.1 Preproduction tests. Unless otherwise specified in the procurement document (see 6.2), preproduction tests on pilor models shall not be required.

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4.3.1.1 Preproduction models. When preproduction tests are specified, (see 6.2) five preproduction models shall be submitted for preproduction model approval and shall be representative of the manufacturer's normal production.

4.3.1.2 Procedure. Preproduction models shall be subjected to the preproduction tests specified in table I in the order shown. All preproduction models shall be subject to the examination and test of group I. The preproduction models shall then be divided and subjected to the tests as specified in group II through IV of Table I.

TABLE I - Preproduction Tests

Tests	Requirement Paragraph	Test Paragraph	Allowable Defects
	GROUP I - 5 Specimens		0
Visual & Mechanical Inspection	3.3 and 3.6	4.4	
Capacity	3.4.1	4.5.1	
Dielectric Withstanding Voltage	3.4.2	4.5.2	
Insulation Resistance	3.4.3	4.5.3	
	GROUP II - 2 Specimens		0
High Temperature	3.5.1	4.5.4.1	
Low Temperature	3.5.2	4.5.4.2	
Temperature Cycling	3.5.3	4.5.4.3	
Humidity	3.5.4	4.5.4.4	
Salt Spray	3.5.5	4.5.4.5	
	GROUP III - 2 Specimens		0
Radic Frequency Interference	3.5.11	4.5.4.6	
Barometric pressure	3.5.6	4.5.4.7	
Shock	3.5.7	4.5.4.8	
Vibration	3.5.8	4.5.4.9	
	GROUP IV - 1 Specimen		0
Fungus	3.5.9	4.5.4.10	
Operational Life	3.5.10	4.5.4.11	

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4.3.1.3 Failure. Failure in excess of the allowable number specified in Table I shall be cause for refusal to grant pilot model approval. Failure is defined as the inability to meet any of the requirements of this specification.

4.3.2 Acceptance inspection and tests. Acceptance inspection and tests shall consist of those tests specified in Table II, in the order shown. Unless otherwise specified in the procurement documents (see 6.2), all items supplied under the specification shall be subjected to acceptance inspection and tests as set forth herein.

TABLE II - Acceptance Inspection and Tests

Test	Requirement Paragraph	Test Paragraph	Allowable Defects
Visual and Mechanical Inspection	3.3 and 3.6	4.4	0
Capacity	3.4.1	4.5.1	0
Dielectric Withstanding Voltage	3.4.2	4.5.2	0
Insulation Resistance	3.4.3	4.5.3	0

4.3.2.1 Inspection lot. An inspection lot shall be as specified in MIL-STD-105. Test data, certified by the contractor shall be submitted in with each lot.

4.3.2.1.1 Lot size. Lot size shall not exceed that quantity of items produced in one continuous operation by the same manufacturer in accordance with the same drawing and specification.

4.3.2.1.2 Lot rejection. Failure of any fan to meet the requirements specified in Table II shall be cause for rejection of the entire lot.

4.4 Visual and mechanical inspection. Fans shall be inspected to verify that the materials, external design and construction, dimensions, weight, marking, and workmanship are as specified in the applicable requirements of 3.3 and 3.6. A visual and mechanical inspection shall also be conducted on fans prepared for delivery to insure compliance to Section 5 requirements.

4.5 Performance tests provisions. The fans shall be subjected to performance testing in accordance with the following.

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4.5.1 Fan and Motor characteristics. The fan shall be tested for air volume and static pressure, as specified in the Standard Test Code for Air Moving Devices approved and adopted by the Air Moving and Conditioning Association (AMCA), Inc. The test procedure shall be in accordance with AMCA fan type and test procedure specified on the detail drawing. All results shall be corrected to standard air as specified in 4.2.2. The motor power requirements shall be tested for conformance to the power requirements specified on the detail drawing.

4.5.2 Dielectric withstanding voltage. The fan shall be tested for dielectric withstanding voltage as specified in method 301, of MIL-STD-202, with the following test voltage applied for a period of 30, plus or minus 15 seconds; 1000 volts ac, 60 hzps applied between the insulated terminals tied together and the motor housing. Any subsequent test for dielectric withstanding voltage (during environmental testing) shall be at 75 percent of the above test voltage.

4.5.3 Insulation resistance. Insulation resistance shall be measured as specified in method 302, condition B, MIL-STD-202. Measurements shall be made between the insulated terminals tied together and the motor housing.

4.5.4 Environmental and service condition tests.

4.5.4.1 High temperature. High temperature test shall be in accordance with MIL-STD-810, method 501, and the following. The fan shall be operated at rated voltage in an ambient temperature of 71° C (160° F) for a period of at least six hours after the temperature has stabilized. The fan shall operate for a period of one hour with a voltage variation of 90% to 120% at the above temperature. At the completion of the test, the fan shall be tested in accordance with 4.5.1.

4.5.4.2 Low temperature. Low temperature test shall be in accordance with MIL-STD-810, method 502, and the following. The fan shall be maintained in an ambient temperature of -54° C (-65° F) for a period of at least six hours after the temperature has stabilized. At the end of the six hour period, and before removal from the test chamber, the rated voltage shall be applied. The fan shall attain the rated speed in accordance with 3.4.1 in five seconds or less. At the completion of the test and at 25° C (77° F) the fan shall be tested in accordance with 4.5.1.

4.5.4.3 Temperature-cycling. The fan shall be temperature cycled as specified in method 102A, condition D, of MIL-STD-202, except that in test condition D, the maximum temperature shall be 71° C (160° F). At the completion of the test the fan shall be tested in accordance with 4.5.1.

4.5.4.4 Humidity. The fan shall be tested for humidity as specified in method 103A, condition A, of MIL-STD-202. Following this test, the fan shall attain rated speed in accordance with 3.4.1 in five seconds or less and shall be tested in accordance with 4.5.1 and 4.5.3.

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4.5.4.5 Salt Spray. The fan shall be tested for salt spray as specified in method 101A, condition B, of MIL-STD-202, using a five percent salt solution. At the conclusion of the salt spray test, the fan shall be examined for marking and finish requirements of 3.3 and measurements shall be made in accordance with 4.5.1 and 4.5.3, in order specified.

4.5.4.6 Radio frequency interference. The fan shall be tested as specified in MIL-STD-461.

4.5.4.7 Barometric pressure. When mounted by its normal mounting means, the fan shall be tested as specified in method 105B of MIL-STD-202, except that the pressures shall be from 17 psia to a pressure equivalent to 10,000 feet altitude. While at high pressure and low pressure, the fan shall be tested in accordance with 4.5.2 and 4.5.3.

4.5.4.8 Shock. The shock test shall be in accordance with MIL-STD-810, method 516, and the following. The fan shall be mounted by its normal mounting means and be subjected to two impact shocks of 15 g in each direction along the mutually perpendicular axis (total eight shocks), one axis to be along the motor armature shaft axis, the other perpendicular to the motor amature shaft axis. The shock impulses shall have a time duration of 11 milliseconds, in a sine waveform. At the conclusion of this test, the fan shall be tested in accordance with 4.5.2, 4.5.3, and 4.5.1, in order specified.

4.5.4.9 Vibration. When mounted by its normal mounting means, the fan shall be tested as specified in method 204A, condition A, of MIL-STD-202, except that the frequency shall be 5 to 500 to 5 Hz and the double amplitude shall be 0.125 inch below 40 Hz. The fan shall be vibrated six times in each of three mutually perpendicular directions. At the conclusion of the test, the fan shall be tested in accordance with 4.5.2. 4.5.3, and 4.5.1, in the order specified.

4.5.4.10 Fungus. The fan shall be tested as specified in the fungus resistance test of MIL-STD-810, method 508. At the conclusion of the test, the fan shall be tested in accordance with 4.4, 4.5.1, and 4.5.3. The supplier's certification of the use of fungus-inert materials may be accepted in lieu of this test.

4.5.11 Operational life. The fan shall be life tested for the length of time and in an ambient temperature as specified on the detail drawing. The fan shall operate at the rated voltage and zero static pressure (free delivery). At the conclusion of the test the fan shall be tested in accordance with 4.5.2 4.5.3, and 4.5.1, in the order specified.

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4.5.4.12 Noise. The fan shall be tested in accordance with MIL-STD-1472 to meet the requirements of 3.5.12.

5. PACKAGING.

5.1 Preservation. Preservation shall be Level A, B, or commercial (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Each fan shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Each fan shall be dried in accordance with MIL-P-116, process D-4.

5.1.1.3 Preservative Application. Exposed bare surface will be preserved by applying preservative conforming to MIL-C-16173, Grade 4. Preserved areas shall be intimately wrapped with barrier material conforming to MIL-B-121, Grade A.

5.1.1.4 Unit Packaging. Fans shall be individually packaged method IIB in accordance with specification MIL-P-116. Each fan cleaned, dried and preserved as defined herein shall be inserted and immobilized in a container conforming to PPP-B-636, domestic class, style RSC. A quantity of desiccant conforming to specification MIL-D-3464, computed in accordance with MIL-P-116 shall be secured within the container and insulated from contact with the exposed metal parts of the fan. The container shall be closed with tape in accordance with specification PPP-T-76. A desiccant indicator conforming to MS-2000-3 shall be secured on the outer top side of the closed container. The fan shall be inserted in a barrier bag conforming to specification MIL-B-131 Class 1 and heat sealed. The bagged fan shall then be inserted in a container conforming to PPP-B-636, style RSC, Class WR completely lined with material conforming to PPP-C-850 of sufficient thickness to prevent the fan from sustaining more than 15G shock (as the specified limit) when dropped under the conditions specified in 3.5.7 and 4.5.4.8. Closure shall be by using tape conforming to PPP-T-76.

5.1.2 Level B. Level B packaging shall be the same as Level A, except omit desiccant and indicator. Method shall be IA-14.

5.1.3 Commercial. Packaging shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, B, or commercial as specified (see 6.2).

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5.2.1 Level A. A specified quantity of fans, packaged as prescribed herein, shall be inserted in a snug-fitting container conforming to PPP-B-601, Style A or B, overseas type. Closure shall be in accordance with the requirements set forth in the appendix of the container specification.

5.2.2 Level B. Level B packing shall be the same as A, except that domestic type container shall be utilized.

5.2.3 Commercial. Packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Level A or B. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. Fans furnished to this specification are intended to be used within the ground support equipment of a guided missile system.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Title, number, and revision level of the applicable detail drawing.

c. Part number

d. Type classification

e. When preproduction model tests are required (see 4.3.1)

f. Disposition of test units of (e) above, when applicable

g. Disposition of rejected parts

h. Applicable levels of preservation, packaging, and packing (see 5.1, 5.2, and 5.3)

i. Environmental measurement tolerance (see 4.2.5)

j. Acceptance inspection and tests (see 4.3.2)

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Air Force - 99
Army - MI
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Review Activities:
Army - EL
Navy - AS
Air Force - 14
User Activities:
Navy - AS, MC

Preparing Activity
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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-F-46851B, FANS, GUIDED MISSILE, GROUND SUPPORT EQUIPMENT GENERAL SPECIFICATION FOR

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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